

People Analytics

Groupwork Use Cases 5 – Recruitment and Selection Analytics

In this final workshop, you will apply analytics techniques to recruitment and selection data to identify potential biases, evaluate the fairness of selection decisions, and assess whether assessment centre data can predict performance and turnover. You will use chi-square tests, logistic and linear regression models to uncover hidden patterns and build evidence-based recommendations.

Prompt 1: Exploring Diversity in the Applicant Pool

Dataset: recruitment_applicants_2023.xlsx

Tasks:

- Recode gender and UG variables.
- Generate frequency tables for Gender, UG, Shortlisted, Interviewed, Offered, Accepted, Joined.
- Interpret patterns in diversity at each recruitment stage.

Deliverables:

- Table of frequencies.
- Initial diagnosis: Where in the pipeline are UG or female applicants being lost?

Prompt 2: Testing for Bias in Shortlisting and Offers

Dataset: recruitment_applicants_2023.xlsx

Tasks:

- Use cross-tabulation and chi-square tests to compare:
 - Gender × Shortlisted
 - UG × Shortlisted
 - Gender × Offer
 - UG × Offer
- Identify significant imbalances.

Deliverables:

- Chi-square test results with interpretation.
- Summary: Are the recruitment decisions fair and unbiased?

Prompt 3: Predicting Shortlisting Using Demographics

Dataset: recruitment_applicants_2023.xlsx

Tasks:

- Run logistic regression with Gender and UG as predictors and ShortlistedNY as target.
- Interpret odds ratios and model fit (Nagelkerke R^2).
- Comment on potential compound bias (e.g., female & UG status).

Deliverables:

- Model output.
- Odds ratios and statistical significance.
- Recommendations to improve fairness in shortlisting.

Prompt 4: Validating Selection Tools Using Performance Ratings

Dataset: selection.xlsx

Tasks:

- Predict Year 1 performance rating using demographic, personality, competency, aptitude, and onboarding variables.
- Use linear regression with standardized coefficients.
- Interpret model fit and variable importance.

Deliverables:

- Regression summary with top predictors.
- Evaluation of selection validity: Which tools truly predict performance?

Prompt 5: Predicting Early Turnover

Dataset: selection.xlsx

Tasks:

- Run logistic regression to predict LeaverYr2.
- Use demographics, assessments, onboarding, and performance rating.
- Evaluate odds ratios and risk factors.

Deliverables:

- Model output and interpretation.
- Recommendations for reducing graduate attrition.